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AGILE PROJECTS DONE RIGHT: LESSONS FROM PROJECT BETTERPRODUCT

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Abstract

Software development projects have evolved over time and newer methods, Agile software development methods are changing the ways software is developed. Agile methods are based on a set of values and activities that reflect the values and teaching Agile methods should discuss both the values as well as the methods themselves. This teaching case describes a successful Agile software product development project. The case describes how Agile software methods, communication tools and prototypes were used to ensure efficient communication and collaboration amongst stakeholders. The case illustrates how artefacts and tools can be used to clarify the project goals and maintain effective customer collaboration. The case also illustrates how the artefacts and tools can be used to mitigate risks of communicational breakdowns. The teaching notes suggest that the case can be used to teach the concepts of Agile software development, stakeholder theory and concept of boundary objects.

Keywords: Agile Software Development, Case study, Stakeholder theory, Boundary objects

1 INTRODUCTION

1.1 The Trend Towards Agile

Software development projects are in midst of transitioning from more traditional, plan-driven development approaches to fast-paced and light-weight endeavours that challenge the development team and involved stakeholders alike. This is mainly due to the immense popularity of Agile software development methods that have grown to be prevalent in contemporary organisations (Hummel 2014). With this development, it is important for students of the IT and IS disciplines to familiarise themselves with the values of the Agile Manifesto, as well as all the different development methods that based upon the approach (Beck et al. 2001).

There is no single way to conduct an Agile software projects as the Agile development is an umbrella term for a collection of different methods. The common thread in all methods is to ensure fast pace of development and flexible project outcomes. The development work is planned out in detail only for the near future. The longer-term project vision is more vague and adaptable for new ideas. Agile project development methods, such as Scrum (Schwaber 2004) guide practitioners by suggesting how to organise the development teams, engage stakeholders and even what kind of physical environment should the teams be located in order to achieve the best results. The multiplicity of the methods and configurations can portray Agile development as very nebulous which is why a single case study is a good starting point for understanding how the projects can successfully apply the methods.

We argue that a good way to learn Agile values and how they affect development work is to delve into an in-depth case study where the project team has embodied the Agile ideals, and executed the project aligned with those values in an exemplary manner. The case we present here illustrates a project that seems to be simple on the surface, but further analysis reveals that even in small-scale software development projects, there are multiple stakeholders involved and a wide variety of tools that have to be utilised in order to inform and manage the stakeholders. This case project also provides an opportunity to identify and shed light on the most common risks associated with the software project development environment, as well as the means available to an organisation to mitigate these risks.

1.2 Case Background

The case project introduced in this paper tells the story of Project BetterProduct, a successful eCommerce product development project that effectively utilised Agile development methods and a variety of complementary artefacts. The project was conducted by the client organisation Escapade (a pseudonym), and their vendor Carmine (a pseudonym) in 2014 and 2015.

1.3 About Escapade Inc

Escapade is an Australian online retailer. They offer products mainly to retail customers but they also have business-to-business clients. The retail sales are done through Escapade's website, a site created and maintained by their own internal IT development team. The business-to-business sales are done via separate IT product, an eCommerce platform, also created by the IT team of Escapade.

The company was founded in the early 2000s and it has grown to become an organisation with approximately eighty employees with over two million online sales transactions. In addition to the IT development team, Escapade consists of a business-to-business client department, a customer service department, a marketing department, a finance department and the management team.

The IT team of Escapade was conducting their projects with a combination of the Scrum ([Schwaber 2004](#)) and Kanban methods ([Anderson 2010](#)). Both methodologies follow the values of Agile Manifesto but the methods have been created to address different aspects of software development projects.

Scrum is focused on the project management side of software development. In Scrum, the development team has a role called Scrum Master who ensures that the Agile principles are followed by the Scrum Team during the development. Another important role is the Product Owner; who provides the development team with the product requirements. The development team, the Scrum Master and the Product Owner conduct short meetings every day called daily Scrums. In these meetings the product status is discussed and the team can clarify things with the Product Owner. In addition to daily meetings, the team organizes regular, recurring meetings where the results of each product iteration are presented to the Product Owner and other customers.

Kanban is a work stream management technique focusing on timely and non-wasteful project delivery. It has its roots in the visual representations of product status as conventionally indicated in Japanese product manufacturing. Each project task is written on a card (physical or virtual) and allocated in Kanban board with swim lanes, which describe the different statuses of the tasks. A simple board could have for example following swim lanes: 'not allocated', 'in progress' or 'done'. The tasks move from lane to lane according to their status. Anyone who is familiar with the board format can easily identify how much of the project is in progress and how much is ready.

1.4 About Carmine Co

Carmine Co is a software consulting firm that conducts user experience design and prototyping of websites, application and website development as well as maintenance and support. The main offering of Carmine are projects that are conducted using the object oriented, open source development framework, Ruby on Rails.

Carmine does not have physical offices. Their employees work from home or are located in the customer's premises. Carmine prides itself as one of the largest Ruby on Rails development companies in Australia. They place a strong emphasis on the experience of their employees as well as the quality of their project outcomes.

For the project with Escaped, Carmine applied the same Agile methodology of project management, Scrum ([Schwaber 2004](#)). The lack of physical office premises meant that Carmine developers had to structure their work around virtual daily stand-up meetings with the customers whilst working from home. Carmine development team had adopted the Scrum roles such as Scrum Master and the Scrum Team albeit the team in the project was somewhat small. The work was broken into smaller requirements called user stories that were stored in a backlog, also a Scrum concepts for requirements management. The implementation order of the stories was decided by the customers and was extensively discussed in a number of meetings, virtual and physical.

2 PROJECT BETTERPRODUCT

2.1 Why Project BetterProduct?

Project BetterProduct was initiated by Escapade to replace an older, existing eCommerce application that was used by the company in order to manage customer and transaction information. The users of the product could conduct purchases of various products via the application based on rewards they were given by their employers. Not only was Escapade using the older system internally, but they had also sold the product as a configurable package to a number of larger organisations that used it as part

of their internal human resources processes. The technology that was used to create the older system was now out-dated and its functionalities were limited. Project BetterProduct would result in a system that is more scalable and flexible, and it is expected to attract new business-to-business customers.

Escapade's management decided that the most feasible way to update the product was to develop a completely new system. Despite the fact that Escapade had an internal, technically proficient IT team, the team lacked the capacity to take on another large project. Internal IT team was already working on other significant improvements to the retail site. After evaluating a number of options, Escapade's management came to the conclusion that they had to find an external software vendor to create the product. They set out to define the product and to find a vendor that would suit the project needs.

2.2 Initiating Project BetterProduct

The Escapade management team for the project consisted of Cate, the Project Sponsor, Bob, the Escapade Chief Technology Officer (CTO) and Dave, the Product Owner. Prior to vendor selection, Dave had gathered an initial list of requirements for the new product. Dave and Cate had written a thorough business proposal based on these initial ideas. Dave and Cate had collated preliminary ideas for the product features that were collected from the members of the organisation who used the older product as part of their daily activities. Dave then formatted the requirements into short stories of how the users might use the said features. The detailed list of the requirements was stored in a spreadsheet, where as the business proposal outlined the high-level view of the requirements, the project budget and a preliminary project schedule. The timeliness and accuracy of the ideas was of the essence, as Cate, the Project Sponsor, explained:

"We went with Agile methods because we did not want to sit there, waiting for nine months for them to deliver an end product, we wanted something now." – Cate, Escapade Project Sponsor

The Escapade management approved the business proposal and the project was subsequently green lighted. Bob, the CTO, began the search for a suitable vendor that would provide Ruby on Rails services and utilise Agile development methods. Cate took a step back from the project's daily operations, and Dave was appointed to be responsible for the vendor communication and project delivery along with Bob.

Escapade selected the vendor, Carmine, based on their Agile project development track record and references. Escapade's managers also revealed that the vendor was chosen partially because of their cultural fit with their organisation. Bob especially had strong opinion of what Agile development constituted of and he wanted that the chosen vendors would align with his view.

"You find a lot of vendors who have no sense of quality and no sense on Agile. When I was speaking to vendors it was easy for me to pick out the ones that were [not truly Agile but just using Agile as a selling point.] Just selling me "I know we're Agile" so I would ask them: "describe your way of Agile" and if their being Agile was to move faster, I knew they were not for us. Saying things such as "we can change things whenever you want", that's not Agile". –Bob, Escapade CTO

The business proposal created by the Product Owner Dave served as a base for the project contract between Escapade and Carmine. Both parties agreed that the contract had to employ Agile development methods, which also meant that there could be changes in the requirements or priorities that were yet unknown. In order to maintain the flexibility, the contract was written with minimum details. Carmine's invoicing was based on the hours their development teams used for the project. Chip explained that Carmine was able to convince their clients into such arrangements because they had good reputation and they were seen as trustworthy vendors.

"We like to say that the client can walk away anytime... When they walk away they've got the software delivered because we are delivering on day-to-day basis... It's a tricky business, it's one of the trickiest things as negotiating a new job with Agile is that the client needs to be very trusting in order to accept that you are not going to rip them off by getting paid upfront. And when you tell them that

you are going to deliver an unknown amount of work, they can sort of get a bit uncomfortable then. So it's taken the company its reputation as able to deliver. If we did not have that reputation it would be much harder to sell." –Chip, Carmine Project Manager

After the contract negotiations, Bob expressed that he was very satisfied with the way the contract was written. He had previous experience of working in small start-up companies that had agreed on too many things they were able to deliver. The projects ended up with the start-up going into administration. Bob wanted to avoid same issues with Carmine and he was pleased that neither organisation was tightly locked by the contract.

2.3 The Commencement of Project BetterProduct

The core team established for the project was small but consisted of members from both organisations, which is in line with the prescriptions of Scrum (see Schwaber 2004). The Carmine side of the project team consisted of Project Manager Chip, Visual Designer Tina, User Experience Designer Natasha and two developers, Jack and Kyle. Chip, the Carmine Project Manager was responsible for customer communication and finances. Kyle, a very experienced developer, filled the role of the project Scrum Master, facilitating the daily Scrum meetings and overseeing the technical implementation of the project. The main customer contact person for the development team was the Product Owner Dave but Bob, the CTO, was also updated weekly on the project's status. The other stakeholders from Escapade were mainly communicating with Dave and Bob. Dave kept the Project Sponsor, Cate, informed on the main project milestones, and Cate, in turn, would then report to Escapade's management.

The Escapade project team members were all co-located in their office but Carmine had members working from home. For example, the Project Manager Chip and the Visual Designer Tina were located in different states while Kyle, the Scrum Master, was actually located in Canada for a significant period of time during the development of the first version of the product. In order to mitigate the issues that time differences might have caused, Kyle worked during the night to match his hours with the other project members. In addition to Kyle's working hours accommodation, Jack, the other developer, worked occasionally in the Escapade office alongside with the Escapade Product Owner Dave. This arrangement facilitated fluent communication between the developers and Dave.

Once the project contract was signed and the project was officially initiated, the vendor developers began to set up the project development environments. An Operations Manager from Escapade's internal IT team oversaw the setup. The project environments included code development, testing and system production servers, which were hosted by an external software-as-a-service cloud provider. The environments were pieced together with automation tools that enabled the process of continuous systems releases for customer review.

2.4 The Discovery Phase

Once the environments were set up, product development could begin. Carmine called the first part of the project the '*Discovery phase*'. Chip, the Carmine Project Manager explained that in the Discovery phase, the project goals were further refined and the requirements were analysed by the developers and a user experience designer. As the first step of the Discovery phase, the User Experience Designer Natasha held workshops with the Escapade Project Sponsor Cate and the Product Owner Dave, in which the preliminary product ideas were discussed. Next, Natasha created a wireframe mock-up of the system and conducted a usability session with a few participants from Escapade. The usability testing participants were selected by Dave based on their availability and skills. Natasha asked the testers to conduct tasks with the wireframes and tell her how they felt about the current ideas.

After these test sessions, she created a new revised version of the wireframe. An example of one of the pages of the wireframes is presented in Figure 1. Escapade's management agreed that this representation was good enough to base the development on and the developers began to write the

code for the actual product. Simultaneously, Tina, the Visual Designer, began to create the look and the feel of the site, based on the wireframes and the requests from Escapade marketing team.

“She took the wireframes and the developers started to work on features already by the time when she got into the project. She designed what the product should look like. We collaborated a lot. Normally with Tina, the visual designer and me, there was always a lot of back and forth. I showed her the wireframes and she told me ‘Yeah, okay. I can do that.’ or ‘That won’t work that well.’ so if she disagreed, I changed the wireframes” -Natasha, the User Experience Designer

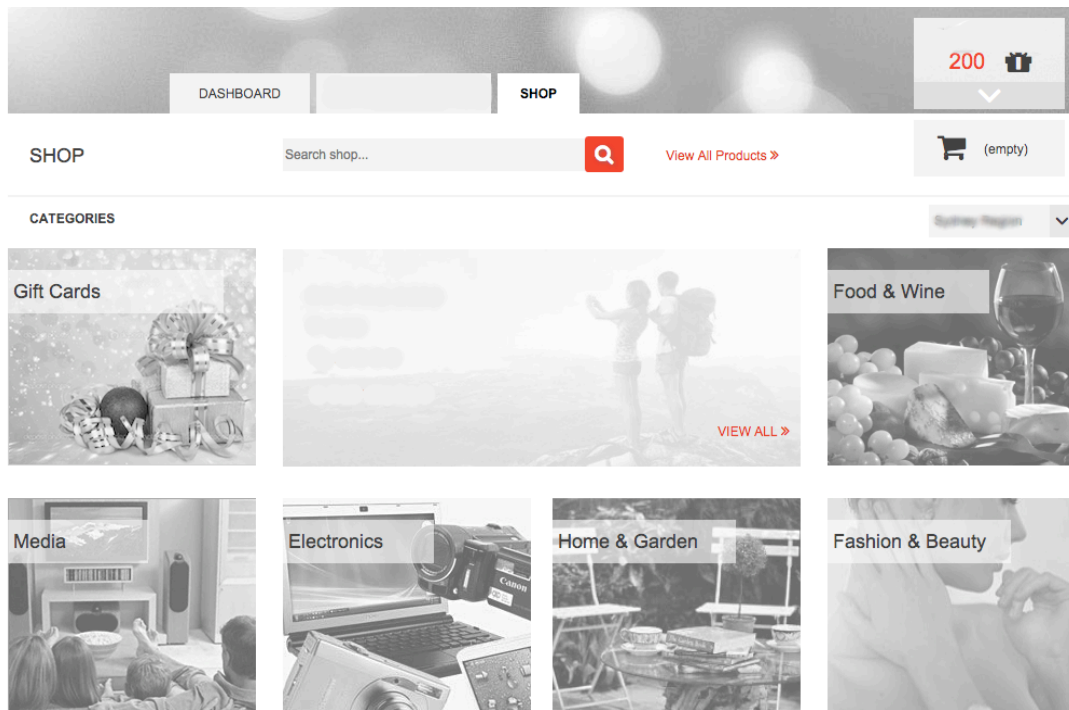


Figure 1. Wireframes page example

2.5 The Development Phase

Product development was conducted in short, two-week iterations, or Sprints as they are called in Scrum methodology. During each Sprint, the team would work on a prioritised set of requirements that were stored in a backlog tool, Pivotal Tracker. In the end of each Sprint, the development team would estimate how many requirements they would be able to finish in the next Sprint. Based on these estimations the requirements were allocated in the Sprints, in priority order. The requirements were written in the form of User Stories, describing the functionality of each requirement in laypersons terms, with no technical jargon. The User Stories were moved across the backlog dashboard according to their statuses that could be roughly categorised as “Waiting”, “In Development”, or “Ready”. The backlog tool was built specifically for Agile project development. The tool’s functionalities supported the framing of user requirements in User Story format, and the Product Owner or the developers could organise the requirements according to their planned Sprints and reorganise them if necessary.

Each morning, the project team would have a daily stand up sessions where they discussed what was completed, what was going to be done and any potential issues. These meetings were held via Skype. There was also an online communications tool, called Flowdock that was used the outside regular meetings:

“There are many forms of communication that we have. In terms of running with our general Scrumban type of methodology, we have our daily stands ups, we have prioritising every week, we

have retros{pective meetings}. In addition we communicate normally via an online chat tool. We find it is fantastic for what we do. Especially keen on the fact that all the developers, external developer team is located all over the world.” –Dave, Escapade Product Owner

The chat tool that was used for communication was also linked to the backlog tool and the flow tools that described the status of the environments. The chat tool users got notifications if automated tests passed or failed. The project team was utilising both the group discussion and one-on-one discussion functionalities of the chat tool, and they had different chat rooms for internal discussions and discussions with Dave, the Product Owner. All the discussions in the tool were logged and stored in the tool itself.

“So Flowdock is for our daily communication. We are able to chat to each other about different aspects of the system and we can break them down by thread. It takes out the face-to-face communication. Pivotal Tracker is for our backlog management and our general priorities management as well. But we do put communication into Flowdock because it is just easier to reference to a feature than within Pivotal Tracker. Very specific feature communication is kept in Pivotal [Tracker]. Where as with Flowdock it's more about conceptual conversation. So that's how we use those two tools and it works brilliantly for us. –Dave, Escapade Product Owner

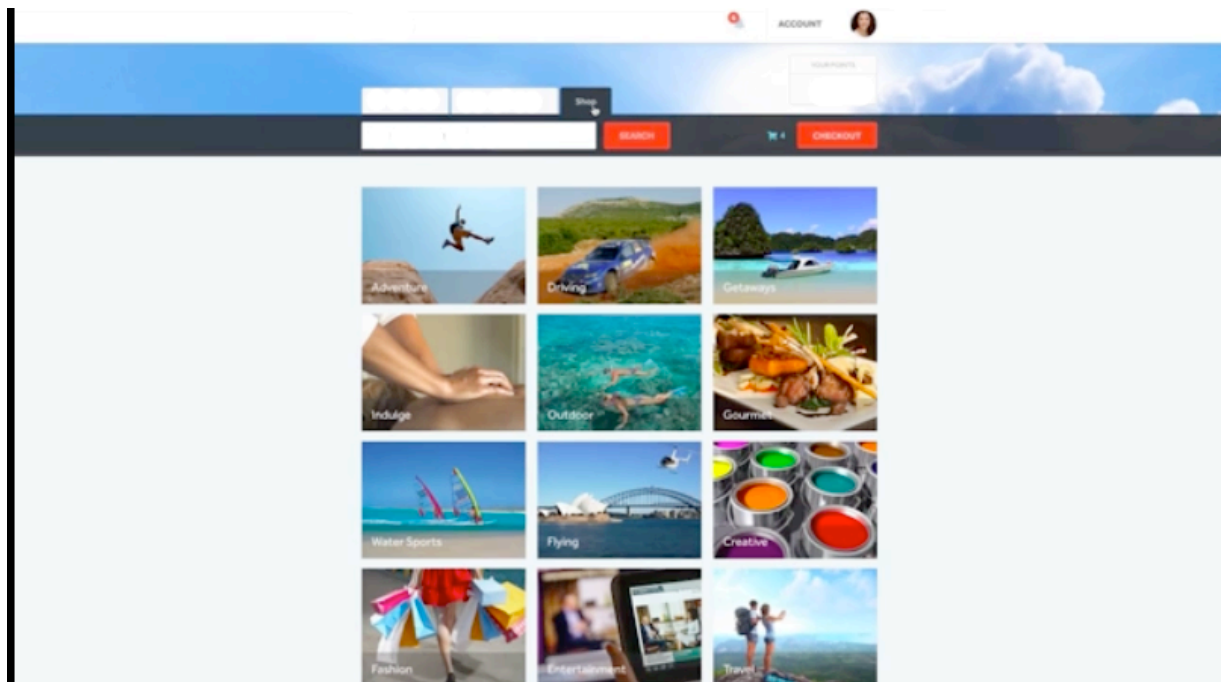


Figure 2. The landing pages of the first released version of the product

After a few months of development, the product was ready to be tested by the Escapade stakeholders. Dave and the Project Sponsor Cate selected a dozen of testers based on their experience with the older system and testing procedures. These testers recorded their findings in an excel spreadsheet that was communicated to the developers by Dave via the backlog tool. After the testing and the subsequent modifications that eventuated from the results, the system was officially released as a Beta version for the whole of Escapade to use internally. The landing page of the first released version is shown in Figure 2.

The development of the remaining requirements continued whilst Escapade stakeholders were using the Beta release and commenting on their user experience at the same time. The feedback that stemmed from those comments was again communicated to the development team through the

backlog tool as a set of requirements or changes to existing requirements. Figure 3 outlines the project schedule and the main milestones.

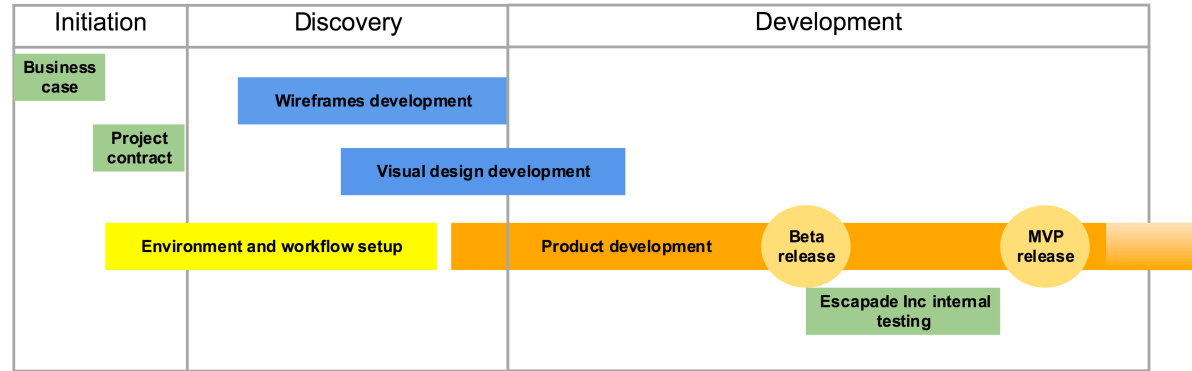


Figure 3. Project schedule and main milestones

2.6 Managerial Communication

The Product Owner Dave was a crucial link between the Escapade stakeholders and the project team. In addition to collaborating on the requirements on a daily basis with the Carmine, he updated the project status for the Project Sponsor, sent out a company wide newsletter that specified the project highlights and upcoming changes, as well as maintained a visual board that showed the project status in the office premises on a weekly basis. Many at Escapade and Carmine praised Dave as a very effective project manager. They said that they always knew that they could contact him in case of emergent issues but seldom had to, because Dave had kept them informed so well.

“Dave has a weekly one-on-one with me on Mondays. So we will go through where he is up to with the project. He will show me the product as is and walk me through the new features that's been released. What's coming up, what's next and this is what we've got to get done.” –Cate, Escapade Project Sponsor

“I think we are communicating really well. That's I think is one of the strong things with Escapade. Our whole communication is one of the most important parts of the project so. Dave is probably talking with my team right now. They are always talking. It is good, we've got very good communication on this project.” –Chip, Carmine Project Manager

Bob, the CTO, was less involved in the day-to-day discussions surrounding the project. There were only a handful of occasions where Bob had to intervene, like when the Project Sponsor felt that some of the requirements were not given the priority they should have from a marketing perspective. Bob was granted the access to the chat room where Dave and the developers discussed project details but he seldom participated in the discussion.

“Thing for my level, if I get too much into the detail, with every project. I have major projects that I manage and I get a bit overwhelmed so that's why Dave is really good. I have a lot of one on one meeting with Dave every week. We go through how he is tracking, how he feels how he is going, any major obstacles, anything I can help with...I usually, we just have a little chat about how it's going, workshop what are we trying to release, what are the main goals..” –Bob, Escapade CTO

Chip, the Project Manager from Carmine made sure that development got done but he was only working on the project part time. The developers and the designers themselves handled much of the customer communications without Chip's involvement. Chip was also keeping Bob up to date on budgeting issues and reacted when the Escapade stakeholders had some concerns or issues.

"I communicate with Bob quite a bit but it is less frequently. It is more like a weekly basis. I'll give him updates every month on how much budget has been spent and he also raises issues. If he's got any concerns he can contact me and likewise I can contact him. But so far this project has a very smooth process so there hasn't been any big issues to raise." –Chip, Carmine Project Manager.

2.7 Closing the Project

The Beta release was not yet ready to be launched to external business-to-business customers, and it took a few more months of work before the Escapade managers decided that the product was ready for external clients to use and test. They selected a few of their business-to-business clients to test the system in its Minimum Viable Product stage. One of the clients was a very large organisation with special technological requirements that were originally unexpected, but these changes were dealt with in the same way as any other new requirements: They were added to the backlog, prioritised and implemented in the next version of the system.

Subsequent to the launch of the Minimum Viable Product, the development of BetterProduct continued but the development team was scaled down from two developers to a single part-time developer. After the changes that were required for the first business-to-business clients were implemented, the next product version was marketed to other business clients. The product is now sold as a Software-as-a-Service offering and now has multiple business clients using it as part of their daily work.

The main deliverable of Project BetterProduct was produced within an acceptable timeframe and it was not over budget. Escapade stakeholders were satisfied with the end results and the product has been well received in the markets. Overall, the project was deemed to be successful from both the financial and user acceptance perspectives. Only time will tell if the product will live up to Escapade's expectations in the long run.

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TEACHING NOTES

1 TEACHING OBJECTIVES

The case of Project BetterProduct can be used to illustrate a project that closely adheres to Agile development guidelines when teaching courses on software development or the management of distributed projects. The conventional means of teaching software project management tends to spend a disproportionate amount of the time discussing the planning and initiation of software projects while glossing over the actual development process (for example: [Pressman \(2005\)](#)).

Depending on the nature of the course, one or more of the following teaching objectives should be emphasized:

- To illustrate the complexity of software development even in the context of a small and fairly straightforward project
- To highlight the means of overcoming the challenges of distributed team communication
- To describe the various stakeholders and their contributions to the project, and provide an introduction to Stakeholder Theory ([Freeman 2010](#))
- To teach a concept of Boundary Objects (Star & Griesemer 1989) and how they can be used to support collaboration between different stakeholders

1.1 Possible Discussion Questions

1. Identify the main project stakeholders and project tools and artefacts that were used during the project. Use a diagram to illustrate the stakeholders and create a table listing the tools and artefacts that facilitated project development in the case study. Identify in the table what each tool/artefact is used for. Discuss why each stakeholder, tool or artefact is important.
2. Is the project too reliant on Product Owner Dave? What actions were taken to mitigate issues if Dave would suddenly leave the project, if any?
3. Compare Agile Manifesto values with the values embodied by Project BetterProduct. Do you think the project implementation described in the case adheres to Agile values?
4. Analyse the tools and artefacts that were used to support development. How do they support Agile development values? Which of these artefacts and tools can be identified as boundary objects and how can they be categorised?

1.2 Teaching Plan

1.2.1 Identify the main project stakeholders and project tools and artefacts that were used during the project. Use a diagram to illustrate the stakeholders and create a table listing the tools and artefacts that facilitated project development in the case study. Identify in the table what each tool/artefact is used for. Discuss why each stakeholder, tool or artefact is important.

Even a small software development project can be complex. There are multiple stakeholders, tools and artefacts described in the case narrative.

The student should analyse the description and identify the main project stakeholders. Donaldson and [Preston \(1995\)](#) define stakeholders: “(a) Stakeholders are persons or groups with legitimate interests in procedural and/or substantive aspects of corporate activity. Stakeholders are identified by their

interests in the corporation, whether the corporation has any corresponding functional interest in them. (b) The interests of all stakeholders are of intrinsic value. That is, each group of stakeholders merits consideration for its own sake and not merely because of its ability to further the interests of some other group, such as the shareowners.”

In order to present the stakeholders in a simple and clear, the students could draw a diagram representing the stakeholders from both client and vendor organisations and also the additional stakeholders such as the cloud service provider and the business-to-business clients of the Escapade. Figure 1 is an example diagram of the project BetterProduct main stakeholders.

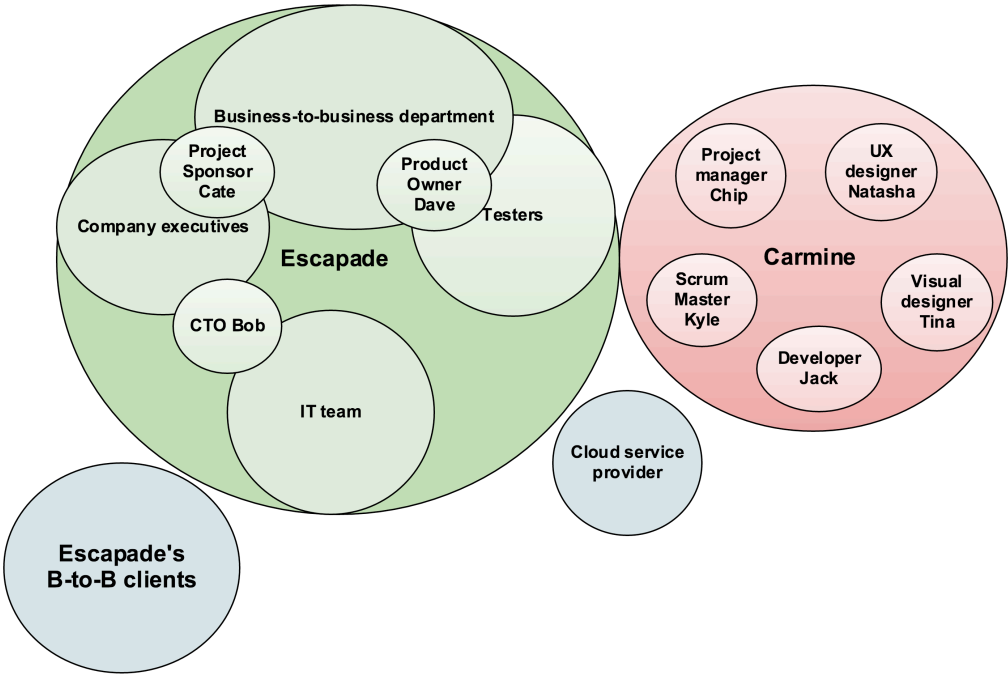


Figure 1 Main project stakeholders

The project artefacts and tools are easier to analyse in conjunction with the project milestones. The project milestones and selected project artefacts, tied to the milestones, are already provided in a form of Figure 3 in the case description (this figure is replicated below as Figure 2). Both the case description and the figure describe how different milestones and main activities took place over the course of the project.

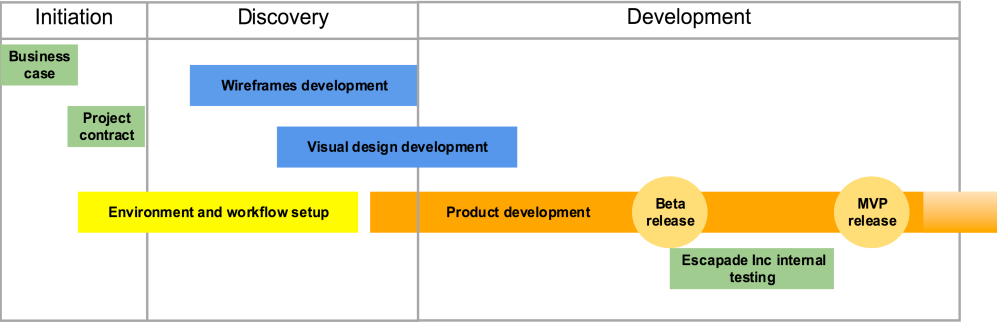


Figure 2. Project schedule and main milestones

Students should be able to list the tools and the artefacts that are described in the narrative but they will have to do a deeper analysis of the quotes and the description in order to identify the use. Student might need to familiarise themselves with the concepts of IT architecture from a textbook on software projects, for example Pressman (2005).

The case description provides two examples of the main artefacts, a screenshot of one of the pages in the wireframes in Figure 1 and a screenshot of the actual product in Figure 2.

Tools and artefacts	Used for
Business proposal: Schedule and budget, initial list of requirements in an Excel spread sheet	Used to establish a baseline of the requirements, schedule and budget. It was also used to convince Escapade management that project BetterProduct was feasible. Used for gathering requirements for the new product version from Escapade 's internal product users and external business customers
Project contract	Used to establish a formal project relationship between the companies
Project development environments: Development, Test, Production	Used to enable product development, testing and use.
Visual design	Used to define the visual look of the product and to accommodate branding and marketing needs
Wireframes	Used to communicate what the product could look like and what could the main functionalities be. They were also used to test the usability of the proposed solution and to obtain user feedback.
Requirements in Pivotal Tracker	Used for capturing requirements as user stories, to visually display the status of each requirement e.g. are they ready, in development, waiting etc. It was also used to provide a platform for testing feedback and comments from Escapade
Chat tool, Flowdock and Skype	Used to discuss the requirements and issues.
Escapade internal testing excel spread sheet	Used to capture the findings from the Escapade user testing.
The product: Beta release, B-to-B client release	Used for communicating efficiently what the product looked like and what were the main functionalities up to date,. It was also used for confirming the designs with the customers and to get feedback from user testing.

Table 1. Tool and artefacts and their use

1.2.2 Is the project too reliant on Product Owner Dave? What actions were taken to mitigate issues if Dave would suddenly leave the project, if any?

Dave is certainly a very central figure in the project. Had Dave suddenly left the project, many of the communication channels would be temporarily severed and key information would be harder to access. Student should discuss what it means to have key people in projects and how their roles can be distributed more evenly across the project team as suggested in Agile methods, for example Scrum, that discuss self-organizing teams.

There were several mitigation protocols implemented in the project. Dave, even though important for the project communication, shared frequently valuable information to other stakeholders. Bob and Chip were both partial to the information shared between Dave and the development team. Cate was kept updated so she was aware of the broad strokes of the project. The backlog tool and the communication tool kept a log of what was discussed regarding the requirements. A replacement for Dave could have a hard time to find all the information but it would be available if one would knew where to look.

1.2.3 Compare Agile Manifesto values with the values embodied by Project BetterProduct. Do you think the project implementation described in the case adheres to Agile values?

Students should first familiarise themselves with the concept of Agile development and Agile values defined in the Agile Manifesto (Beck et al. 2001). A good way to reflect on Agile values is to compare the Scrum process and with single pass waterfall model (Royce 1970).

Table lists few examples of the practices that were applied by the project team in order to follow the values of the Manifesto. Students should discuss what other actions could have been taken and how do they evaluate the values.

Agile value (Beck et al. 2001).	Practices used in BetterProduct project
Individuals and interactions over processes and tools	The scrum meetings, daily and weekly. Chat communication between the organisations, Scrum roles that emphasise shared responsibilities (scrum master, product owner)
Working software over comprehensive documentation	Prototype to verify the requirements was developed. Multiple releases of the software. Customers got a working version of the system very early in the project.
Customer collaboration over contract negotiation	Both parties loosely wrote the contract. Communication about finances and schedule was facilitated between the parties via weekly meetings.
Responding to change over following a plan	New requirements that emerged from testing and use were taken into consideration via normal Sprint process.

Table 2. Agile values and Agile practices used in the case project

1.2.4 Analyse the tools and artefacts that were used to support development. How do they support Agile development values? Which of these artefacts and tools can be identified as boundary objects and how can they be categorised?

The concept of boundary objects will aid students when conducting a deeper analysis of the tools and artefacts used in the project. Boundary objects are objects that are used to convey information across boundaries of groups or organization. Boundary objects are objects that reside in the boundaries of organisations and people. They are plastic and adaptable to local needs but also robust enough to maintain common identity (Star & Griesemer 1989). Boundary objects allow different groups to work together without consensus (Star 2010).

Students should discuss where the boundaries of the projects can be drawn in this case study. The boundary can be drawn between the Agile development team and the non-Agile Escapade stakeholders but other possible boundaries can be also identified. Students should justify why they assume a boundary might exist. An example categorisation of the boundaries and boundary objects in Table 3 is adopted from Zaitsev et al. (2016, forthcoming). Students should analyse how the objects are used in order to create common ground across the boundary they have defined.

Boundary objects	Object use to mitigate boundaries
Infrastructure boundary objects: Contract	Facilitate the existence of other objects, support the chosen methodology
Process boundary objects: Feedback notes in a spreadsheet	Store documentation, feedback and discussion data Loosely linked with other objects, information overlap
Projective boundary objects: Wireframes, Visual design The software system	Reflect and project the project goals Provide input for customer feedback and new requirement ideas

Table 3. Agile values and boundary objects used in the case project

1.3 Suggested Readings

- Beck, K., Beedle, M., van Bennekum, A., Cockburn, A., Cunningham, W., Fowler, M., Grenning, J., Highsmith, J., Hunt, A., and Jeffries, R. (2001). "The Agile Manifesto." available at <http://www.agilemanifesto.org>
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- Star, S. L., and Griesemer, J. R. (1989). Institutional ecology, translations' and boundary objects: Amateurs and professionals in Berkeley's Museum of Vertebrate Zoology, 1907-39. *Social studies of science*, 19 (3), p387-420.
- Zaitsev A., Tan B. and Gal U. (2016, forthcoming), *Collaboration Amidst Volatility: the Evolving Nature of Boundary Objects in Agile Software Development*, *Proceedings of the European Conference on Information Systems (ECIS)*. 2016.